name	region	year	pop_n (in thousands)	pop_n (Millions)	ARC_n [1]	ARC_r [2]	ARC_u [3]	ARC_n_full	ARC_r_full	ARC_u_full	ARC_diff [4]
Afghanistan	South Asia	2015			2.7503	2.667882638	2.668211856				-0.0003
Afghanistan	South Asia	2020	38928.33984	38.92833984							
Albania	Europe & Central Asia	2015			0.3347	0.69281669	0.043748432				0.6491
Albania	Europe & Central Asia	2020	2877.800049	2.877800049							
Algeria	Middle East & North Africa	2015			0.2056	0.33696621	0.102884232				0.2341
Algeria	Middle East & North Africa	2020	43851.04297	43.85104297							
American Samoa	East Asia & Pacific	2015			0.0309	Null	Null	Full access			Null
American Samoa	East Asia & Pacific	2020	55.1969986	0.055196998 6							
Andorra	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000
Andorra	Europe & Central Asia	2020	77.26499939	0.07726499939							
Angola	Sub-Saharan Africa	2015			0.5702	0.218771442	0.3046367				-0.0859
Angola	Sub-Saharan Africa	2020	32866.26953	32.86626953							
Anguilla	Latin America & Caribbean	2015			0.0000	Null	0				Null
Anguilla	Latin America & Caribbean	2017	14.5880003	0.0145880003							
Antigua and Barbuda	Latin America & Caribbean	2015			0.0000	Null	Null				Null
Antigua and Barbuda	Latin America & Caribbean	2017	95.42500305	0.09542500305							
Argentina	Latin America & Caribbean	2015			Null	Null	0.05365152			Full access	Null
Argentina	Latin America & Caribbean	2020	45195.77734	45.19577734							
Armenia	Europe & Central Asia	2015			0.0837	0.189044146	0.022125564	Full access		Full access	0.1669
Armenia	Europe & Central Asia	2020	2963.233887	2.963233887							
Aruba	Latin America & Caribbean	2015			0.0000	Null	Null				Null
Aruba	Latin America & Caribbean	2016	104.8649979	0.1048649979							
Australia	East Asia & Pacific	2015			0.0000	0	0	Full access	Full access	Full access	0.0000
Australia	East Asia & Pacific	2020	25499.88086	25.49988086							
Austria	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000
Austria	Europe & Central Asia	2020	9006.400391	9.006400391							
Azerbaijan	Europe & Central Asia	2015			0.7246	1.356516936	0.145769594				1.2107
Azerbaijan	Europe & Central Asia	2020	10139.1748	10.1391748							
Bahamas	Latin America & Caribbean	2015			0.0000	Null	Null				Null
Bahamas	Latin America & Caribbean	2019	389.4859924	0.3894859924							
Bahrain	Europe & Central Asia	2015			0.0000	Null	Null	Full access			Null
Bahrain	Europe & Central Asia	2020	1701.583008	1.701583008							
Bangladesh	South Asia	2015			0.1193	0.221445296	-0.065600342				0.2870
Bangladesh	South Asia	2020	164689.3906	164.6893906							
Barbados	Latin America & Caribbean	2015			0.0080	Null	Null				Null
Barbados	Latin America & Caribbean	2020	287.3710022	0.2873710022							
Belarus	Europe & Central Asia	2015			0.0115	0.046690142	0.016367688				0.0303
Belarus	Europe & Central Asia	2020	9449.321289	9.449321289							
Belgium	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000
Belgium	Europe & Central Asia	2020	11589.61621	11.58961621							
Belize	Latin America & Caribbean	2015			0.2453	0.468275104	-0.025266				0.4935
Belize	Latin America & Caribbean	2020	397.6210022	0.3976210022							
Benin	Sub-Saharan Africa	2015			0.1255	0.253583104	-0.207828504				0.4614
Benin	Sub-Saharan Africa	2020	12123.19824	12.12319824							
Bermuda	North America	2015			0.0000	Null	0	Full access		Full access	Null
Bermuda	North America	2020	62.27299881	0.06227299881							
Bhutan	South Asia	2015			0.2184	0.31336614	0.041858342				0.2715
Bhutan	South Asia	2020	771.6119995	0.7716119995							
Bolivia (Plurinational State of)	Latin America & Caribbean	2015			0.5856	1.239292526	0.186424248				1.0529
Bolivia (Plurinational State of)	Latin America & Caribbean	2020	11673.0293	11.6730293							
Bosnia and Herzegovina	Europe & Central Asia	2015			-0.0092	0	0				0.0000
Bosnia and Herzegovina	Europe & Central Asia	2020	3280.814941	3.280814941							
Botswana	Sub-Saharan Africa	2015			0.7535	1.430238006	0.2164756				1.2138
Botswana	Sub-Saharan Africa	2020	2351.625	2.351625							
Brazil	Latin America & Caribbean	2015			0.3035	1.43784627	0.103474058				1.3344
Brazil	Latin America & Caribbean	2020	212559.4063	212.5594063							
British Virgin Islands	Latin America & Caribbean	2015			0.0000	Null	Null	Full access			Null
British Virgin Islands	Latin America & Caribbean	2020	30.23699951	0.03023699951							
Brunei Darussalam	East Asia & Pacific	2015			0.0780	Null	0	Full access		Full access	Null
Brunei Darussalam	East Asia & Pacific	2020	437.4830017	0.4374830017							
Bulgaria	Europe & Central Asia	2015			-0.0370	-0.122970474	-0.016507464			Full access	-0.1065
Bulgaria	Europe & Central Asia	2020	6948.444824	6.948444824							
Burkina Faso	Sub-Saharan Africa	2015			-0.5845	-1.22738096	0.049225276				-1.2766
Burkina Faso	Sub-Saharan Africa	2020	20903.27734	20.90327734							
Burundi	Sub-Saharan Africa	2015			0.5251	0.425499672	0.360134654				0.0654
Burundi	Sub-Saharan Africa	2020	11890.78125	11.89078125							
Cabo Verde	Sub-Saharan Africa	2015			0.6653	1.456442266	0.130977098				1.3255
Cabo Verde	Sub-Saharan Africa	2020	555.9879761	0.5559879761							
Cambodia	East Asia & Pacific	2015			0.5552	0.47120783	0.381977146				0.0892
Cambodia	East Asia & Pacific	2020	16718.9707	16.7189707							
Cameroon	Sub-Saharan Africa	2015			0.3493	0.304382382	-0.03466461				0.3390
Cameroon	Sub-Saharan Africa	2020	26545.86328	26.54586328							
Canada	North America	2015			-0.0015	0.048860164	-0.013240238				0.0621
Canada	North America	2020	37742.15625	37.74215625							
Cayman Islands	Latin America & Caribbean	2015			0.0000	Null	0				Null
Cayman Islands	Latin America & Caribbean	2016	62.56399918	0.06256399918							
Central African Republic	Sub-Saharan Africa	2015			-1.0218	-0.756962728	-1.620086812				0.8631
Central African Republic	Sub-Saharan Africa	2020	4829.76416	4.82976416							
Chad	Sub-Saharan Africa	2015			0.3577	0.399262902	-0.112285738				0.5115
Chad	Sub-Saharan Africa	2020	16425.85938	16.42585938							
Channel Islands	Europe & Central Asia	2015			0.0000	Null	Null				Null
Channel Islands	Europe & Central Asia	2017	168.6660004	0.168666004							
Chile	Latin America & Caribbean	2015			0.0987	0.5864603	0.028128604	Full access		Full access	0.5583
Chile	Latin America & Caribbean	2020	19116.20898	19.11620898							
China	East Asia & Pacific	2015			0.4997	1.036409118	-0.08095426				1.1174
China	East Asia & Pacific	2020	1463140.5	1463.1405							
China, Hong Kong SAR	East Asia & Pacific	2015			0.0000	Null	0	Full access		Full access	Null
China, Hong Kong SAR	East Asia & Pacific	2020	7496.987793	7.496987793							
China, Macao SAR	East Asia & Pacific	2015			0.0000	Null	0	Full access		Full access	Null
China, Macao SAR	East Asia & Pacific	2020	649.34198	0.64934198							
Colombia	Latin America & Caribbean	2015			0.2312	0.698669394	0.057696086			Full access	0.6410
Colombia	Latin America & Caribbean	2020	50882.88281	50.88288281							
Comoros	Sub-Saharan Africa	2015			0.0196	0	0				0.0000
Comoros	Sub-Saharan Africa	2019	850.8909912	0.8508909912							
Congo	Sub-Saharan Africa	2015			0.5270	0.866515448	0.0599182				0.8066
Congo	Sub-Saharan Africa	2020	5518.091797	5.518091797							
Cook Islands	East Asia & Pacific	2015			0.0036	Null	Null	Full access			Null
Cook Islands	East Asia & Pacific	2020	17.56399918	0.0175639991 8							
Costa Rica	Latin America & Caribbean	2015			0.0864	0.292043502	0.02238755			Full access	0.2697
Costa Rica	Latin America & Caribbean	2020	5094.11377	5.09411377							
Côte d'Ivoire	Sub-Saharan Africa	2015			-0.0333	-0.024144972	-0.311478412				0.2873
Côte d'Ivoire	Sub-Saharan Africa	2020	26378.27539	26.37827539							
Croatia	Europe & Central Asia	2015			Null	Null	0			Full access	Null
Croatia	Europe & Central Asia	2020	4105.268066	4.105268066							
Cuba	Latin America & Caribbean	2015			0.2262	0.788318096	0.054655054				0.7337
Cuba	Latin America & Caribbean	2020	11326.61621	11.32661621							
Curaçao	Latin America & Caribbean	2015			0.0000	Null	Null				Null
Curaçao	Latin America & Caribbean	2017	161.9859924	0.1619859924							
Cyprus	Europe & Central Asia	2015			-0.0047	-0.003084764	-0.005496838	Full access	Full access	Full access	0.0024
Cyprus	Europe & Central Asia	2020	1207.360962	1.207360962							
Czech Republic	Europe & Central Asia	2015			0.0001	0	0	Full access	Full access	Full access	0.0000
Czech Republic	Europe & Central Asia	2020	10708.98242	10.70898242							
Democratic People's Republic of Korea	East Asia & Pacific	2015			-0.2742	-0.572668708	-0.115002716				-0.4577
Democratic People's Republic of Korea	East Asia & Pacific	2020	25778.81445	25.77881445							
Democratic Republic of the Congo	Sub-Saharan Africa	2015			0.6468	0.272055708	0.436595398				-0.1645
Democratic Republic of the Congo	Sub-Saharan Africa	2020	89561.40625	89.56140625							
Denmark	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000
Denmark	Europe & Central Asia	2020	5792.203125	5.792203125							
Djibouti	Sub-Saharan Africa	2015			0.0526	-0.295694432	0.092757016				-0.3885
Djibouti	Sub-Saharan Africa	2020	988.0020142	0.9880020142							

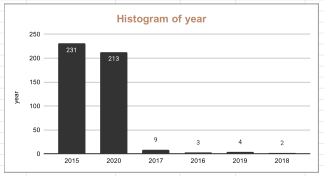
Investigating year representation

Min year	2015
Max year	2020
Average y_diff [5	4.8
Min y_diff	1
Max y_diff	5
Count y_diff = 0	0

The minimum gap is 1 year, so the average is much larger than the minimum (meaning some countries have observations close together, but that's rare).

The maximum gap is 5 years, and the average (4.80) is very close to the maximum, meaning the dataset is mostly recorded every 5 years for most countries.

The data collection frequency is mainly 5-year intervals, with a few cases where a country has data 1 year apart.



The dataset is not evenly distributed across time. Most country observations are recorded in 2015 and 2020, suggesting that the dataset mainly captures measurements at roughly five-year intervals (baseline vs latest). The small counts for 2016–2019 indicate a limited number of countries with intermediate-year data, so this is not a full annual time series for most countries.

Investigating Annual Rates of Change (ARC)

	ARC_n [6]	ARC_r [7]	ARC_u [8]
average	0.2767	0.484493696	0.1548234028
minimum	-1.0218	-1.22738096	-1.620086812
maximum	2.7503	2.667882638	2.668211856

A lower ARC average (as calculated before) does not necessarily indicate less progress in changing access to basic water services because it takes into account countries that already have 100% access.

Investigating access by area

We are counting the string "Null" occurrences in those ARC columns.

Nb of countries that have missing ARC values	
ARC_n [9]	2
ARC_r [10]	64
ARC_u [11]	50

Calculating the number of countries that have full access per population

The number of "full access" occurrences per area	
National	62
Rural	29
Urban	55

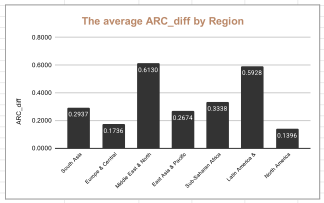
Many countries have 100% urban or national access, which forces ARC to 0 and can pull averages down.

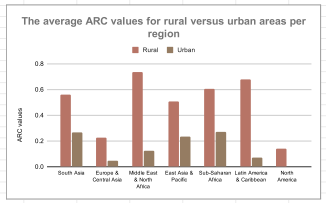
Nb of countries that have 'ARC = 0' and that don't already have full access	
National	16
Rural	5
Urban	7

Nb of countries that have 'ARC > 0' and that don't already have full access	
National	135
Rural	116
Urban	93

Nb of countries that have 'ARC < 0' and that don't already have full access	
National	16
Rural	17
Urban	26

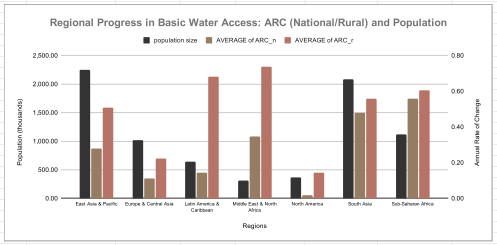
Checking that we've considered all countries for the conditions	231 [12]
	231 [13]





Investigating access by region

Regions	Nb untique countries	pop_n (Millions)	AVERAGE of ARC_n	AVERAGE of ARC_r	AVERAGE of ARC_u
East Asia & Pacific	40	2,247.54	0.28	0.51	0.23
Europe & Central Asia	64	1,017.48	0.11	0.22	0.05
Latin America & Caribbean	48	642.51	0.14	0.68	0.07
Middle East & North Africa	10	311.07	0.35	0.74	0.12
North America	5	368.87	0.02	0.14	0.00
South Asia	11	2,082.32	0.48	0.56	0.27
Sub-Saharan Africa	53	1,124.03	0.56	0.60	0.27
Grand Total	**231**	**7,793.82**	**0.2767**	**0.484493696**	**0.1548234028**



Dominica	Latin America & Caribbean	2015			0.0000	Null	Null				Null	
Dominica	Latin America & Caribbean	2017	71.45999908	0.07145999908								
Dominican Republic	Latin America & Caribbean	2015			0.1175	0.218684118	0.011664354					0.2070
Dominican Republic	Latin America & Caribbean	2020	10847.9043	10.8479043								
Ecuador	Latin America & Caribbean	2015			0.4592	0.8996526	0.17353216					0.7261
Ecuador	Latin America & Caribbean	2020	17643.06055	17.64306055								
Egypt	Middle East & North Africa	2015			0.0667	0.106191446	0.013872274			Full access		0.0923
Egypt	Middle East & North Africa	2020	102334.4063	102.3344063								
El Salvador	Latin America & Caribbean	2015			0.4769	1.098636152	0.140890432					0.9577
El Salvador	Latin America & Caribbean	2020	6486.201172	6.486201172								
Equatorial Guinea	Sub-Saharan Africa	2015			0.2436	0	0					0.0000
Equatorial Guinea	Sub-Saharan Africa	2017	1262.008057	1.262008057								
Eritrea	Sub-Saharan Africa	2015			0.3846	0	0					0.0000
Eritrea	Sub-Saharan Africa	2016	3376.558105	3.376558105								
Estonia	Europe & Central Asia	2015			-0.0321	Null	0	Full access		Full access	Null	
Estonia	Europe & Central Asia	2020	1326.53894	1.32653894								
Eswatini	Sub-Saharan Africa	2015			0.8098	0.868292768	0.361032584					0.5073
Eswatini	Sub-Saharan Africa	2020	1160.16394	1.16016394								
Ethiopia	Sub-Saharan Africa	2015			1.5091	1.515674352	0.450730324					1.0649
Ethiopia	Sub-Saharan Africa	2020	114963.5859	114.9635859								
Faeroe Islands	Europe & Central Asia	2015			0.0000	Null	Null	Full access			Null	
Faeroe Islands	Europe & Central Asia	2020	48.86500168	0.04886500168								
Falkland Islands (Malvinas)	Latin America & Caribbean	2015			0.0978	0	0			Full access		0.0000
Falkland Islands (Malvinas)	Latin America & Caribbean	2020	3.48300004	0.00348300004								
Fiji	East Asia & Pacific	2015			0.0077	-0.058100668	-0.021760122					-0.0363
Fiji	East Asia & Pacific	2020	896.4439697	0.8964439697								
Finland	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Finland	Europe & Central Asia	2020	5540.717773	5.540717773								
France	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
France	Europe & Central Asia	2020	65273.51172	65.27351172								
French Guiana	Latin America & Caribbean	2015			0.0440	Null	Null				Null	
French Guiana	Latin America & Caribbean	2020	298.6820068	0.2986820068								
French Polynesia	East Asia & Pacific	2015			0.0000	Null	Null	Full access			Null	
French Polynesia	East Asia & Pacific	2020	280.9039917	0.2809039917								
Gabon	Sub-Saharan Africa	2015			0.2967	0.27117117	0.09824146					0.1729
Gabon	Sub-Saharan Africa	2020	2225.728027	2.225728027								
Gambia	Sub-Saharan Africa	2015			0.3460	0.186726128	0.243046602					-0.0563
Gambia	Sub-Saharan Africa	2020	2416.664063	2.416664063								
Georgia	Europe & Central Asia	2015			0.3187	0.595346722	0.077637698					0.5177
Georgia	Europe & Central Asia	2020	3989.175049	3.989175049								
Germany	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Germany	Europe & Central Asia	2020	83783.94531	83.78394531								
Ghana	Sub-Saharan Africa	2015			1.1266	0.936889056	0.995247936					-0.0584
Ghana	Sub-Saharan Africa	2020	31072.94531	31.07294531								
Gibraltar	Europe & Central Asia	2015			0.0000	Null	0	Full access		Full access	Null	
Gibraltar	Europe & Central Asia	2020	33.69100189	0.03369100189								
Greece	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Greece	Europe & Central Asia	2020	10423.05566	10.42305566								
Greenland	North America	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Greenland	North America	2020	56.77199936	0.05677199936								
Grenada	Latin America & Caribbean	2015			0.0000	Null	Null				Null	
Grenada	Latin America & Caribbean	2017	110.8740005	0.1108740005								
Guadeloupe	Latin America & Caribbean	2015			0.0384	Null	Null	Full access			Null	
Guadeloupe	Latin America & Caribbean	2020	400.1270142	0.4001270142								
Guam	East Asia & Pacific	2015			0.0000	Null	Null	Full access			Null	
Guam	East Asia & Pacific	2020	168.7830048	0.1687830048								
Guatemala	Latin America & Caribbean	2015			0.3745	0.575314724	0.117436106					0.4579
Guatemala	Latin America & Caribbean	2020	17915.56641	17.91556641								
Guinea	Sub-Saharan Africa	2015			0.0693	-0.24019527	0.286836476					-0.5270
Guinea	Sub-Saharan Africa	2020	13132.79199	13.13279199								
Guinea-Bissau	Sub-Saharan Africa	2015			0.0505	0.28183505	-0.471876662					0.7537
Guinea-Bissau	Sub-Saharan Africa	2020	1967.998047	1.967998047								
Guyana	Latin America & Caribbean	2015			0.0786	0.101144158	0			Full access		0.1011
Guyana	Latin America & Caribbean	2020	786.559021	0.786559021								
Haiti	Sub-Saharan Africa	2015			0.4256	0.08585641	-0.009011568					0.0949
Haiti	Sub-Saharan Africa	2020	11402.5332	11.4025332								
Honduras	Latin America & Caribbean	2015			0.4870	0.68774516	0.208424026					0.4793
Honduras	Latin America & Caribbean	2020	9904.608398	9.904608398								
Hungary	Europe & Central Asia	2015			0.0059	0.019982056	0	Full access	Full access	Full access		0.0200
Hungary	Europe & Central Asia	2020	9660.349609	9.660349609								
Iceland	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Iceland	Europe & Central Asia	2020	341.25	0.34125								
India	South Asia	2015			0.4703	0.642100556	0.053794338					0.5883
India	South Asia	2020	1380004.375	1380.004375								
Indonesia	East Asia & Pacific	2015			0.7776	1.007434238	0.427604062					0.5798
Indonesia	East Asia & Pacific	2020	273523.625	273.523625								
Iran (Islamic Republic of)	South Asia	2015			0.1351	0.333246546	0.03014517					0.3031
Iran (Islamic Republic of)	South Asia	2020	83992.95313	83.99295313								
Iraq	Middle East & North Africa	2015			0.8225	2.013687334	0.296175686					1.7175
Iraq	Middle East & North Africa	2020	40222.50391	40.22250391								
Ireland	Europe & Central Asia	2015			0.0055	0.023620912	-0.001211156					0.0248
Ireland	Europe & Central Asia	2020	4937.795898	4.937795898								
Isle of Man	Europe & Central Asia	2015			0.0000	Null	Null				Null	
Isle of Man	Europe & Central Asia	2020	85.03199768	0.085031997 68								
Israel	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Israel	Europe & Central Asia	2020	8655.541016	8.655541016								
Italy	Europe & Central Asia	2015			0.0000	Null	Null	Full access			Null	
Italy	Europe & Central Asia	2020	60461.82813	60.46182813								
Jamaica	Latin America & Caribbean	2015			0.1226	0.179840452	0.021605134					0.1582
Jamaica	Latin America & Caribbean	2020	2961.160889	2.961160889								
Japan	East Asia & Pacific	2015			0.0312	Null	Null				Null	
Japan	East Asia & Pacific	2020	126476.4609	126.4764609								
Jordan	Middle East & North Africa	2015			-0.0133	-0.01016734	-0.018220846					0.0081
Jordan	Middle East & North Africa	2020	10203.13965	10.20313965								
Kazakhstan	Europe & Central Asia	2015			0.0819	0.2383509	-0.045335692					0.2837
Kazakhstan	Europe & Central Asia	2020	18776.70703	18.77670703								
Kenya	Sub-Saharan Africa	2015			0.6872	0.701563404	0.004416706					0.6971
Kenya	Sub-Saharan Africa	2020	53771.30078	53.77130078								
Kiribati	East Asia & Pacific	2015			0.8246	0.61438646	0.55124857					0.0631
Kiribati	East Asia & Pacific	2020	119.4459991	0.1194459991								
Kuwait	Middle East & North Africa	2015			0.0000	Null	Null	Full access			Null	
Kuwait	Middle East & North Africa	2020	4270.562988	4.270562988								
Kyrgyzstan	Europe & Central Asia	2015			0.6515	0.830642272	0.257269172					0.5734
Kyrgyzstan	Europe & Central Asia	2020	6524.190918	6.524190918								
Lao People's Democratic Republic	East Asia & Pacific	2015			1.5518	1.646654574	1.002896448					0.6438
Lao People's Democratic Republic	East Asia & Pacific	2020	7275.556152	7.275556152								
Latvia	Europe & Central Asia	2015			0.0515	0.144900826	0.00720627					0.1377
Latvia	Europe & Central Asia	2020	1886.202026	1.886202026								
Lebanon	Middle East & North Africa	2015			0.2751	Null	Null				Null	
Lebanon	Middle East & North Africa	2020	6825.441895	6.825441895								
Lesotho	Sub-Saharan Africa	2015			0.3261	0.045894334	0.624326482					-0.5784
Lesotho	Sub-Saharan Africa	2020	2142.251953	2.142251953								
Liberia	Sub-Saharan Africa	2015			0.5384	0.567406986	0.314001124					0.2534
Liberia	Sub-Saharan Africa	2020	5057.676758	5.057676758								
Libya	Sub-Saharan Africa	2015			0.5220	Null	Null				Null	
Libya	Sub-Saharan Africa	2020	6871.287109	6.871287109								
Liechtenstein	Europe & Central Asia	2015			0.0000	Null	Null	Full access			Null	
Liechtenstein	Europe & Central Asia	2020	38.13700104	0.038137001 04								
Lithuania	Europe & Central Asia	2015			0.2818	0.612859164	0.10540193					0.5075
Lithuania	Europe & Central Asia	2020	2722.291016	2.722291016								
Luxembourg	Europe & Central Asia	2015			-0.0090	-0.128571428	0	Full access		Full access		-0.1286

Luxembourg	Europe & Central Asia	2020	625.9760132	0.6259760132								
Madagascar	Sub-Saharan Africa	2015			0.8988	0.65236851	0.516035244				0.1363	
Madagascar	Sub-Saharan Africa	2020	27691.01953	27.69101953								
Malawi	Sub-Saharan Africa	2015			0.7786	0.868887024	0.043763672				0.8251	
Malawi	Sub-Saharan Africa	2020	19129.95508	19.12995508								
Malaysia	South Asia	2015			0.0095	-0.14417222	-0.008094028				-0.1361	
Malaysia	South Asia	2020	32365.99805	32.36599805								
Maldives	South Asia	2015			0.1460	0.207862646	0.057636648				0.1502	
Maldives	South Asia	2020	540.5419922	0.5405419922								
Mali	Sub-Saharan Africa	2015			1.6296	1.737036236	1.001209844				0.7358	
Mali	Sub-Saharan Africa	2020	20250.83398	20.25083398								
Malta	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000	
Malta	Europe & Central Asia	2020	441.5390015	0.4415390015								
Marshall Islands	East Asia & Pacific	2015			0.0459	0.030272	0.090284526				-0.0600	
Marshall Islands	East Asia & Pacific	2020	59.19400024	0.05919400024								
Martinique	Latin America & Caribbean	2015			0.0092	Null	Null	Full access			Null	
Martinique	Latin America & Caribbean	2020	375.2650146	0.375265014 6								
Mauritania	Sub-Saharan Africa	2015			0.9415	0.643699344	0.573901212				0.0698	
Mauritania	Sub-Saharan Africa	2020	4649.660156	4.649660156								
Mauritius	Sub-Saharan Africa	2015			0.0000	0	0	Full access	Full access	Full access	0.0000	
Mauritius	Sub-Saharan Africa	2020	1271.766968	1.271766968								
Mayotte	Sub-Saharan Africa	2015			-0.0933	Null	Null				Null	
Mayotte	Sub-Saharan Africa	2020	272.8129883	0.2728129883								
Mexico	Latin America & Caribbean	2015			0.3308	1.01007498	0.14728942				0.8628	
Mexico	Latin America & Caribbean	2020	128932.75	128.93275								
Micronesia (Federated States of)	East Asia & Pacific	2015			0.0000	Null	Null				Null	
Micronesia (Federated States of)	East Asia & Pacific	2019	113.810997	0.113810997								
Monaco	Europe & Central Asia	2015			0.0000	Null	0	Full access		Full access	Null	
Monaco	Europe & Central Asia	2020	39.24399948	0.03924399948								
Mongolia	East Asia & Pacific	2015			0.9365	1.795372626	0.492087668				1.3033	
Mongolia	East Asia & Pacific	2020	3278.291992	3.278291992								
Montenegro	Europe & Central Asia	2015			0.2931	0.440245072	0.211412228				0.2288	
Montenegro	Europe & Central Asia	2020	628.0620117	0.6280620117								
Montserrat	Latin America & Caribbean	2015			0.0013	Null	Null				Null	
Montserrat	Latin America & Caribbean	2020	4.999000072	0.004999000072								
Morocco	Middle East & North Africa	2015			1.3302	2.634720044	0.305466198				2.3293	
Morocco	Middle East & North Africa	2020	36910.55859	36.91055859								
Mozambique	Sub-Saharan Africa	2015			2.4387	2.579061548	1.562736418				1.0163	
Mozambique	Sub-Saharan Africa	2020	31255.43555	31.25543555								
Myanmar	East Asia & Pacific	2015			2.0261	2.23459023	1.390321954				0.8443	
Myanmar	East Asia & Pacific	2020	54409.79297	54.40979297								
Namibia	Sub-Saharan Africa	2015			0.3492	0.22302721	-0.055454054				0.2785	
Namibia	Sub-Saharan Africa	2020	2540.916016	2.540916016								
Nauru	East Asia & Pacific	2015			0.0000	Null	0	Full access		Full access	Null	
Nauru	East Asia & Pacific	2020	10.83399963	0.01083399963								
Nepal	South Asia	2015			0.4532	0.592258402	-0.143407588				0.7357	
Nepal	South Asia	2020	29136.80859	29.13680859								
Netherlands	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000	
Netherlands	Europe & Central Asia	2020	17134.87305	17.13487305								
New Caledonia	East Asia & Pacific	2015			0.2334	Null	Null				Null	
New Caledonia	East Asia & Pacific	2020	285.4909973	0.2854909973								
New Zealand	East Asia & Pacific	2015			0.0000	0	0	Full access	Full access	Full access	0.0000	
New Zealand	East Asia & Pacific	2020	4822.23291	4.82223291								
Nicaragua	Latin America & Caribbean	2015			0.0711	-0.079442242	0.032840346				-0.1123	
Nicaragua	Latin America & Caribbean	2020	6624.554199	6.624554199								
Niger	Sub-Saharan Africa	2015			0.2859	0.364843534	-0.338939856				0.7038	
Niger	Sub-Saharan Africa	2020	24206.63672	24.20663672								
Nigeria	Sub-Saharan Africa	2015			1.7659	1.633152206	1.381769374				0.2514	
Nigeria	Sub-Saharan Africa	2020	206139.5938	206.1395938								
Niue	East Asia & Pacific	2015			-0.1191	Null	Null				Null	
Niue	East Asia & Pacific	2020	1.618000031	0.001618000031								
North Macedonia	Europe & Central Asia	2015			0.0541	-0.032903712	0.116888982				-0.1498	
North Macedonia	Europe & Central Asia	2020	2083.379883	2.083379883								
Northern Mariana Islands	East Asia & Pacific	2015			0.0040	Null	Null	Full access			Null	
Northern Mariana Islands	East Asia & Pacific	2020	57.55699921	0.05755699921								
Norway	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000	
Norway	Europe & Central Asia	2020	5421.242188	5.421242188								
Oman	Middle East & North Africa	2015			0.3859	0.37677429	0.164570984				0.2122	
Oman	Middle East & North Africa	2020	5106.62207	5.10662207								
Pakistan	South Asia	2015			0.1388	0.2897808	-0.155671798				0.4455	
Pakistan	South Asia	2020	220892.3281	220.8923281								
Palau	East Asia & Pacific	2015			0.0055	0.02862271	-0.000099948	Full access	Full access	Full access	0.0287	
Palau	East Asia & Pacific	2020	18.09199905	0.01809199905								
Panama	Latin America & Caribbean	2015			0.2936	0.528700746	0.115643056				0.4131	
Panama	Latin America & Caribbean	2020	4314.768066	4.314768066								
Papua New Guinea	East Asia & Pacific	2015			0.7717	0.816142134	0.234025716				0.5821	
Papua New Guinea	East Asia & Pacific	2020	8947.027344	8.947027344								
Paraguay	Latin America & Caribbean	2015			0.5515	1.153305448	0.158939962				0.9944	
Paraguay	Latin America & Caribbean	2020	7132.529785	7.132529785								
Peru	Latin America & Caribbean	2015			0.5715	1.50236293	0.260691412				1.2417	
Peru	Latin America & Caribbean	2020	32971.84766	32.97184766								
Philippines	South Asia	2015			0.4183	0.570754178	0.210076072				0.3607	
Philippines	South Asia	2020	109581.0859	109.5810859								
Poland	Europe & Central Asia	2015			0.1469	0.292419354	0.050967742			Full access	0.2415	
Poland	Europe & Central Asia	2020	37846.60547	37.84660547								
Portugal	Europe & Central Asia	2015			0.0015	0	0	Full access	Full access	Full access	0.0000	
Portugal	Europe & Central Asia	2020	10196.70703	10.19670703								
Puerto Rico	Latin America & Caribbean	2015			0.1149	Null	Null				Null	
Puerto Rico	Latin America & Caribbean	2020	2860.840088	2.860840088								
Qatar	Europe & Central Asia	2015			-0.0247	Null	Null	Full access			Null	
Qatar	Europe & Central Asia	2020	2881.060059	2.881060059								
Republic of Korea	East Asia & Pacific	2015			0.0917	Null	Null				Null	
Republic of Korea	East Asia & Pacific	2020	51269.18359	51.26918359								
Republic of Moldova	Europe & Central Asia	2015			0.3988	0.587709474	0.123009016				0.4647	
Republic of Moldova	Europe & Central Asia	2020	4033.962891	4.033962891								
Réunion	Sub-Saharan Africa	2015			0.0177	Null	Null	Full access			Null	
Réunion	Sub-Saharan Africa	2020	895.3079834	0.895307983 4								
Romania	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access	0.0000	
Romania	Europe & Central Asia	2020	19237.68164	19.23768164								
Russian Federation	Europe & Central Asia	2015			0.0931	0.28679859	0.011332944				0.2755	
Russian Federation	Europe & Central Asia	2020	145934.4531	145.9344531								
Rwanda	Sub-Saharan Africa	2015			0.6585	0.657951734	0.523632384				0.1343	
Rwanda	Sub-Saharan Africa	2020	12952.20898	12.95220898								
Saint Barthelemy	Latin America & Caribbean	2015			0.0070	Null	0.007016146	Full access		Full access	Null	
Saint Barthelemy	Latin America & Caribbean	2020	9.885	0.009885								
Saint Helena	Sub-Saharan Africa	2015			0.0300	Null	Null				Null	
Saint Helena	Sub-Saharan Africa	2020	6.071000099	0.006071000099								
Saint Kitts and Nevis	Latin America & Caribbean	2015			0.0000	Null	Null				Null	
Saint Kitts and Nevis	Latin America & Caribbean	2017	52.0359993	0.0520359993								
Saint Lucia	Latin America & Caribbean	2015			0.2640	0.300352518	0.102357802				0.1980	
Saint Lucia	Latin America & Caribbean	2020	183.6289978	0.183628998								
Saint Martin (French part)	Latin America & Caribbean	2015			-0.0001	Null	-0.000100562	Full access		Full access	Null	
Saint Martin (French part)	Latin America & Caribbean	2020	38.659	0.038659								
Saint Pierre and Miquelon	North America	2015			0.0000	Null	Null				Null	
Saint Pierre and Miquelon	North America	2020	5.795000076	0.005795000076								
Saint Vincent and the Grenadines	Latin America & Caribbean	2015			0.0000	Null	Null				Null	
Saint Vincent and the Grenadines	Latin America & Caribbean	2018	110.2099991	0.1102099991								
Samoa	East Asia & Pacific	2015			0.1033	0.078781444	0.211708264				-0.1329	
Samoa	East Asia & Pacific	2020	198.4100037	0.1984100037								
San Marino	Europe & Central Asia	2015			0.0000	Null	Null	Full access			Null	
San Marino	Europe & Central Asia	2020	33.93799973	0.0339379973								

Sao Tome and Principe	Sub-Saharan Africa	2015			0.2204	0.60357957	-0.00529962					0.6089
Sao Tome and Principe	Sub-Saharan Africa	2020	219.1609955	0.2191609955								
Saudi Arabia	Europe & Central Asia	2015			0.0904	Null	Null	Full access			Null	
Saudi Arabia	Europe & Central Asia	2020	34813.86719	34.81386719								
Senegal	Sub-Saharan Africa	2015			1.2606	1.860345328	0.354800636					1.5055
Senegal	Sub-Saharan Africa	2020	16743.92969	16.74392969								
Serbia	Europe & Central Asia	2015			0.3818	0.22428914	0.50974024					-0.2855
Serbia	Europe & Central Asia	2020	8737.370117	8.737370117								
Seychelles	East Asia & Pacific	2015			0.1502	Null	Null				Null	
Seychelles	East Asia & Pacific	2019	97.74099731	0.09774099731								
Sierra Leone	Sub-Saharan Africa	2015			1.1838	1.444597766	0.542560016					0.9020
Sierra Leone	Sub-Saharan Africa	2020	7976.984863	7.976984863								
Singapore	East Asia & Pacific	2015			0.0000	Null	0	Full access		Full access	Null	
Singapore	East Asia & Pacific	2020	5850.342773	5.850342773								
Sint Maarten (Dutch part)	Latin America & Caribbean	2015			0.0000	Null	0				Null	
Sint Maarten (Dutch part)	Latin America & Caribbean	2017	41.44400024	0.04144400024								
Slovakia	Europe & Central Asia	2015			0.0001	0	0	Full access	Full access	Full access		0.0000
Slovakia	Europe & Central Asia	2020	5459.643066	5.459643066								
Slovenia	Europe & Central Asia	2015			0.0000	Null	Null	Full access			Null	
Slovenia	Europe & Central Asia	2020	2078.931885	2.078931885								
Solomon Islands	East Asia & Pacific	2015			-0.4058	-0.72137414	0.028760968					-0.7501
Solomon Islands	East Asia & Pacific	2020	686.8779907	0.6868779907								
Somalia	Sub-Saharan Africa	2015			1.5532	1.569534044	0.96851942					0.6010
Somalia	Sub-Saharan Africa	2020	15893.21875	15.89321875								
South Africa	Sub-Saharan Africa	2015			0.4017	0.85914853	0.031450278					0.8277
South Africa	Sub-Saharan Africa	2020	59308.69141	59.30869141								
South Sudan	Sub-Saharan Africa	2015			-0.0593	-0.626630184	1.862165418					-2.4888
South Sudan	Sub-Saharan Africa	2020	11193.72852	11.19372852								
Spain	Europe & Central Asia	2015			-0.0002	0	0	Full access	Full access	Full access		0.0000
Spain	Europe & Central Asia	2020	46754.78125	46.75478125								
Sri Lanka	South Asia	2015			0.4233	0.456065912	0.231426734					0.2246
Sri Lanka	South Asia	2020	21413.25	21.41325								
Sudan	Middle East & North Africa	2015			0.2997	0.298317868	0.137440398					0.1609
Sudan	Middle East & North Africa	2020	43849.26953	43.84926953								
Suriname	Latin America & Caribbean	2015			0.4484	1.198959262	0.062090834					1.1369
Suriname	Latin America & Caribbean	2020	586.6339722	0.5866339722								
Sweden	Europe & Central Asia	2015			0.0091	0.033877642	0.004879458	Full access	Full access	Full access		0.0290
Sweden	Europe & Central Asia	2020	10099.26953	10.09926953								
Switzerland	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Switzerland	Europe & Central Asia	2020	8654.618164	8.654618164								
Syrian Arab Republic	Middle East & North Africa	2015			0.0832	0.13940798	-0.010364566					0.1498
Syrian Arab Republic	Middle East & North Africa	2020	17500.65625	17.50065625								
Tajikistan	Europe & Central Asia	2015			1.1226	1.418283428	0.204333694					1.2139
Tajikistan	Europe & Central Asia	2020	9537.641602	9.537641602								
Thailand	East Asia & Pacific	2015			0.1482	0.224096658	0.064892632			Full access		0.1592
Thailand	East Asia & Pacific	2020	69799.97656	69.79997656								
Timor-Leste	East Asia & Pacific	2015			2.0491	2.305681174	1.237175338					1.0685
Timor-Leste	East Asia & Pacific	2020	1318.442017	1.318442017								
Togo	Sub-Saharan Africa	2015			0.9893	0.975073894	0.492516062					0.4826
Togo	Sub-Saharan Africa	2020	8278.737305	8.278737305								
Tokelau	East Asia & Pacific	2015			0.0501	0.05005613	Null				Null	
Tokelau	East Asia & Pacific	2020	1.350000024	0.001350000024								
Tonga	East Asia & Pacific	2015			0.0363	0.015006062	0.108109526					-0.0931
Tonga	East Asia & Pacific	2020	105.6969986	0.1056969986								
Trinidad and Tobago	Latin America & Caribbean	2015			0.1478	Null	Null				Null	
Trinidad and Tobago	Latin America & Caribbean	2020	1399.490967	1.399490967								
Tunisia	Sub-Saharan Africa	2015			0.5396	1.669111792	-0.013650062					1.6828
Tunisia	Sub-Saharan Africa	2020	11818.61816	11.81861816								
Turkey	Europe & Central Asia	2015			0.1516	0.364257356	0.066586558					0.2977
Turkey	Europe & Central Asia	2020	84339.07031	84.33907031								
Turkmenistan	Europe & Central Asia	2015			0.4682	0.756043368	0.184005656					0.5720
Turkmenistan	Europe & Central Asia	2020	6031.187012	6.031187012								
Turks and Caicos Islands	Latin America & Caribbean	2015			0.0000	Null	Null				Null	
Turks and Caicos Islands	Latin America & Caribbean	2018	37.66699982	0.03766699982								
Tuvalu	East Asia & Pacific	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
Tuvalu	East Asia & Pacific	2020	11.79199982	0.01179199982								
Uganda	Sub-Saharan Africa	2015			1.6027	1.70283729	0.447299142					1.2555
Uganda	Sub-Saharan Africa	2020	45741	45.741								
Ukraine	Europe & Central Asia	2015			0.0543	0.231377678	-0.011247684					0.2426
Ukraine	Europe & Central Asia	2020	43733.75781	43.73375781								
United Arab Emirates	Europe & Central Asia	2015			-0.0011	Null	Null	Full access			Null	
United Arab Emirates	Europe & Central Asia	2020	9890.400391	9.890400391								
United Kingdom	Europe & Central Asia	2015			0.0000	0	0	Full access	Full access	Full access		0.0000
United Kingdom	Europe & Central Asia	2020	67886.00781	67.88600781								
United Republic of Tanzania	Sub-Saharan Africa	2015			1.5506	1.316708156	1.0666139					0.2501
United Republic of Tanzania	Sub-Saharan Africa	2020	59734.21484	59.73421484								
United States of America	North America	2015			0.0873	0.378133766	0.021372932			Full access		0.3568
United States of America	North America	2020	331002.6563	331.0026563								
United States Virgin Islands	Latin America & Caribbean	2015			0.0000	Null	Null				Null	
United States Virgin Islands	Latin America & Caribbean	2020	104.4229965	0.1044229965								
Uruguay	Latin America & Caribbean	2015			0.0729	0.760997232	0.032654492			Full access		0.7283
Uruguay	Latin America & Caribbean	2020	3473.727051	3.473727051								
Uzbekistan	Europe & Central Asia	2015			0.0600	0.086038162	0.039397988					0.0466
Uzbekistan	Europe & Central Asia	2020	33469.19922	33.46919922								
Vanuatu	East Asia & Pacific	2015			0.2486	0.283830114	0.092528802					0.1913
Vanuatu	East Asia & Pacific	2020	307.1499939	0.3071499939								
Venezuela (Bolivarian Republic of)	Latin America & Caribbean	2015			-0.1771	Null	Null				Null	
Venezuela (Bolivarian Republic of)	Latin America & Caribbean	2020	28435.94336	28.43594336								
Viet Nam	East Asia & Pacific	2015			0.7113	0.93629409	0.194059298					0.7422
Viet Nam	East Asia & Pacific	2020	97338.58594	97.33858594								
Wallis and Futuna Islands	East Asia & Pacific	2015			-0.0356	-0.035618834	Null				Null	
Wallis and Futuna Islands	East Asia & Pacific	2020	11.24600029	0.01124600029								
West Bank and Gaza Strip	Europe & Central Asia	2015			0.3001	0.666882814	0.184140962					0.4827
West Bank and Gaza Strip	Europe & Central Asia	2020	5101.416016	5.101416016								
Yemen	Europe & Central Asia	2015			0.9769	1.144613242	0.188323488					0.9563
Yemen	Europe & Central Asia	2020	29825.96875	29.82596875								
Zambia	Sub-Saharan Africa	2015			0.8147	0.922090802	0.165671474					0.7564
Zambia	Sub-Saharan Africa	2020	18383.95508	18.38395508								
Zimbabwe	Sub-Saharan Africa	2015			-0.4577	-0.593526372	-0.134701358					-0.4588
Zimbabwe	Sub-Saharan Africa	2020	14862.92676	14.86292676								

[1] The national Annual Rates of Change (ARC) in %

[2] The rural Annual Rates of Change (ARC) in %

[3] The urban Annual Rates of Change (ARC) in %

[4] The difference between the rural ARC (ARC_r) and urban ARC (ARC_u)

[5] The average year difference across all countries

[6] The national Annual Rates of Change (ARC)

[7] The rural Annual Rates of Change (ARC)

[8] The urban Annual Rates of Change (ARC)

[9] The national Annual Rates of Change (ARC)

[10] The rural Annual Rates of Change (ARC)

[11] The urban Annual Rates of Change (ARC)

[12] No value, full access, ARC = 0, ARC < 0, and ARC > 0

[13] Summing the number of countries.